Synergies Between Nuclear Security and Critical Infrastructure: National Legal and Regulatory Frameworks

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Abstract:

Nuclear reactors and other nuclear facilities should comprise part of a nation's critical infrastructure assets. Key cross-sector interdependencies, in relation to energy, transportation systems, communications, emergency services, water, information technologies and others, result in inevitable synergies between legal frameworks for the security of nuclear facilities and legal frameworks for the protection of critical infrastructure. The protection of nuclear facilities against sabotage and other malicious acts is paramount in ensuring energy security and thus ensuring uninterrupted energy supply. The protection of other sectors, such as uninterrupted communications, secure water supply, and others, supports a safe and secure operation of nuclear facilities. Some countries rely on broader critical infrastructure frameworks to impose security requirements on nuclear facilities, or to achieve robust cybersecurity systems. This paper will analyze the interdependencies and synergies between the legal and regulatory frameworks for critical infrastructure protection and nuclear facilities' security by comparing various national frameworks. The paper will also propose modalities to leverage the best practices and requirements from each framework towards energy security goals and stronger national nuclear security regimes.

I. Introduction

Nuclear reactors and other nuclear facilities should comprise part of a state's critical infrastructure assets; however, States, regional bodies, and international organizations have taken different approaches to defining and regulating critical infrastructure, including integrating elements of nuclear security within broader critical infrastructure frameworks. For instance, the United Nations Office of Disaster Risk Reduction defines critical infrastructure as "the physical structures, facilities, networks, and other assets which provide services that are essential to the social and economic functioning of a community or society."[1] The United States (U.S.) Department of Homeland Security's Cybersecurity and Infrastructure Security Agency (CISA) identifies 16 critical infrastructure sectors and defines critical infrastructure as "those assets, systems, and networks that provide functions necessary to our way of life."[2] The European Union defines critical infrastructure as "an asset, a facility, equipment, a network or a system, or a part of an asset, a facility, equipment, a network or a system, which is necessary for the provision of an essential service."[3]

Nuclear reactors and other nuclear facilities are not expressly included in the previously mentioned regimes' definitions of "critical infrastructure." However, the International Atomic

[1] https://www.undrr.org/terminology/critical-infrastructure
[2] https://www.cisa.gov/topics/critical-infrastructure-security-and-resilience
[3] Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the resilience of critical entities, Article 2.

Energy Agency (IAEA) discusses nuclear facilities in the context of a state's broader critical infrastructure, noting that "facilities handling nuclear material or other radioactive material, and undertaking associated activities, are critical infrastructure which require high levels of safety and security."[4]

The IAEA's Nuclear Security Series (NSS) also refer to critical infrastructure when discussing various aspects of nuclear security.[5] NSS No. 17-T (Rev. 1) explicitly calls for requirements taking the form of national regulations that provide for the development and implementation of a computer security program to ensure that nuclear facilities are protected from malicious acts and sabotage.[6] Additionally, NSS No. 13 provides guidance for states regarding the physical protection of nuclear facilities and nuclear material. A state's physical protection regime should endeavor to protect against the unauthorized removal of nuclear material through theft or other unlawful taking; to locate and recover missing nuclear material; to protect against sabotage; and to mitigate or minimize the effects of sabotage.[7] NSS No. 13 provides guidance to states on developing a legal and regulatory framework for physical protection, the establishment or designation of a competent authority to implement the framework, and details the importance of a risk-based physical protection system and measures.

Critical infrastructure frameworks have complimentary goals to frameworks on nuclear and radiological emergency preparedness and response and nuclear security. Critical infrastructure frameworks aim to ensure the resilience and protection of economies and societies. By seeking to protect nuclear facilities and material, nuclear security frameworks such as the Convention on the Physical Protection of Nuclear Material (CPPNM) and its Amendment also seek to protect people and the environment from the effects of theft, smuggling, sabotage, or other harms. Including nuclear facilities as part of a state's critical infrastructure can help further reduce the risks of economic and societal harm and compliment international agreements relating to nuclear security and emergency preparedness and response.

II. Regional and National Frameworks on Protecting Critical Infrastructure

a. European Union

As an initial matter, the European Union (EU) does not regulate the physical protection of nuclear material and facilities; rather, this responsibility resides with individual Member States.[8]

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[4] https://www.iaea.org/bulletin/what-goes-into-making-a-computer-security-programme

[5] See International Atomic Energy Agency, Objective and Essential Elements of a State's Nuclear Security Regime, IAEA Nuclear Security Series No. 20, Section 3.10; International Atomic Energy Agency, National Nuclear Security Threat Assessment, Design Basis Threats and Representative Threat Statements, IAEA Nuclear Security Series No. 10-G (Rev. 1), Section 5.11 and Section 6.19; International Atomic Energy Agency, Establishing the Nuclear Security Infrastructure for a Nuclear Power Programme, IAEA Nuclear Security Series No. 19, Action 4-4; International Atomic Energy Agency, Risk Informed Approach for Nuclear Security Measures for Nuclear and Other Radioactive Material out of Regulatory Control, IAEA Nuclear Security Series No. 24-G, Section 4.4.

[6] International Atomic Energy Agency, Computer Security Techniques for Nuclear Facilities, IAEA Nuclear Security Series No. 17-T (Rev. 1), IAEA, Vienna (2021)

[7] International Atomic Energy Agency, Nuclear Security Series No. 13, Nuclear Security Recommendations on Physical Protection of Nuclear Material and Nuclear Facilities (INFCIRC/225/Revision 5), Section 2.

[8] Amendment to the Convention on the Physical Protection of Nuclear Material, Article 2. The EU does engage on nuclear security issues at the international level through the High Representative of the Union for Foreign Affairs and Security Policy, as well as through the European External Action Service (EEAS).

The EU has addressed the protection of critical infrastructure, however, beginning in 2008 through its Directive 2008/114/EC of 8 December 2008 on the identification and designation of European critical infrastructures and the assessment of the need to improve their protection focuses on the protection of critical infrastructure in the energy and transport sectors. The Directive defined critical infrastructure as "an asset, system or part thereof located in Member States which is essential for the maintenance of vital societal functions, health, safety, security, economic, or social well-being of people, and the disruption or destruction of which would have a significant impact in a Member State as a result of the failure to maintain those functions."[9] The Directive focused on so-called "European critical infrastructure," or ECIs, defined as critical infrastructure that if disrupted or destroyed, would have a "significant impact" on at least two Member States.[10] Member States could designate infrastructure as an ECI following an agreement between that Member State and other states that would be "significantly affected" by the ECI.[11]

In 2022, the European Union adopted Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the resilience of critical entities, which repealed Directive 2008/114/EC. The preambular text notes that an evaluation of Directive 2008/114/EC carried out in 2019 found that "due to the increasingly interconnected and cross-border nature of operations using critical infrastructure, protective measures relating to individual assets alone are insufficient to prevent all disruptions from taking place. Therefore, it is necessary to shift the approach towards ensuring that risks are better accounted for, that the role and duties of critical entities as providers of services essential to the functioning of the internal market are better defined and coherent, and that Union rules are adopted to enhance the resilience of critical entities."[12] Accordingly, Directive 2022/2557, also called the Critical Entities Resilience Directive (CER), seeks to harmonize national approaches towards the protection of critical infrastructure across the European Union, particularly the national designation of critical entities and the requirements imposed on those entities. It covers eleven sectors, including the energy sector, and activities such as electricity production and energy storage.[13]

Directive 2022/2557 defines critical infrastructure as "an asset, a facility, equipment, a network or a system, or a part of an asset, a facility, equipment, a network or a system, which is necessary for the provision of an essential service."[14] The Directive defines essential service as "a service which is crucial for the maintenance of vital societal functions, economic activities, public health and safety, or the environment."[15] At its core, the Directive requires EU Member States to protect critical infrastructure, which includes nuclear power plants, to ensure the resilience of essential services in the face of dynamic risks.

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[9] Council Directive 2008/114/EC of 8 December 2008 on the identification and designation of European critical infrastructures and the assessment of the need to improve their protection, Article 2(a).
[10] Id. at Article 2(b).
[11] Id. at Article 4(3).
[12] Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the resilience of critical entities, Preamble.
[13] European Commission, "Enhancing EU Resilience: A Step Forward to Identify Critical Entities for Key Sectors," https://ec.europa.eu/commission/presscorner/detail/en/ip_23_3992.
[14] Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the resilience of critical entities, Article 2.
[15] Id.

The Directive requires Member States to develop a national strategy for enhancing the resilience of critical entities, which should "set out strategic objectives and policy measures" to ensure a coherent, efficient, and comprehensive approach to critical infrastructure protection.[16] Member States should use a "risk-based approach" to protecting critical infrastructure and carry out an assessment of the relevant risks to essential services, including "relevant natural and man-made risks, including those of a cross-sectoral or cross-border nature, accidents, natural disasters, public health emergencies and hybrid threats or other antagonistic threats, including terrorist offenses."[17] Member States must designate a national authority responsible for the application and enforcement of the Directive,[18] and Member States should cooperate with each other, particularly regarding critical entities that may affect or have a connection with other states.[19] Member States have until October 17, 2024 to adopt national measures to give effect to the Directive.[20]

In 2022, the European Commission adopted a Directive on cybersecurity to strengthen the protection and resilience of critical infrastructure in the field of cybersecurity. Directive 2022/2555 requires Member States to adopt national cybersecurity strategies and to identify relevant authorities to oversee cybersecurity.[21] Building upon the framework established in an earlier 2016 Directive on cybersecurity, Directive 2022/2555, also called the NIS 2 Directive, seeks to facilitate cooperation and the exchange of information between Member States on cybersecurity.[22] Although the Directive designates certain aspects of energy sector as an "essential entity" needing cybersecurity measures, nuclear facilities are not explicitly mentioned.

b. South Africa

Regarding the protection of critical infrastructure, South Africa passed the Critical Infrastructure Protection Act 8 of 2019.[23] The Act aims to "secure critical infrastructure against threats," "ensure that objective criteria are developed for the identification, declaration, and protection of critical infrastructure," "secure critical infrastructure in the Republic by creating an environment in which public safety, public confidence, and basic public services are promoted through the implementation of measures aimed at securing critical infrastructures and by mitigating risks to critical infrastructures through assessment of vulnerabilities and the implementation of appropriate measures."[24]

The Act provides that inspectors, police officials who possess "an appropriate security clearance certificate," have "at least the rank of a warrant officer," and who are "experienced in infrastructure protection," will be designated by the National Commissioner.[25] Inspectors can

[16] Id. at Article 4.
[17] Id. at Article 5.
[18] Id. at Article 9.
[19] Id. at Article 11.
[20] Id at Article 26.
[21] Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 on measures for a high common level of cybersecurity across the Union, amending Regulation (EU) No 910/2014 and Directive (EU) 2018/1972, and repealing Directive (EU) 2016/1148 (NIS 2 Directive), Article 7-9.
[22] Id. at Article 14.
[23] Critical Infrastructure Protection Act of 2019, https://static.pmg.org.za/Critical_Infra_Protection_Act8of2019.pdf.
[24] Id. at Section 2.
[25] Id. at Section 10.

conduct inspections of critical infrastructure "at any reasonable time," review physical security assessments, preserve confidentiality of all matters concerning operational activities of critical infrastructure, and order any persons in control of critical infrastructure to take steps to comply with the Act.[26]

The Act defines infrastructure as "any building, center, establishment, facility, installation, pipeline, premises or systems needed for the functioning of society, the Government or enterprises of the Republic, and includes any transport network or network for the delivery of electricity or water."[27] Infrastructure can be designated critical if "the functioning of such infrastructure is essential for the economy, national security, public safety, and the continuous provision of basic public services," and where "the loss, damage, disruption, or immobilization of such infrastructure may severely prejudice the functioning or stability of the Republic; the public interest with regard to safety and the maintenance of law and order; and national security."[28] A "person in control of infrastructure" can submit an application for that infrastructure to be designated as critical to the National Commissioner of the South African Police Service, who then conducts a "physical security assessment" of the infrastructure to verify information in the application and determine whether the infrastructure should be considered low-risk, medium-risk, or high-risk.[29] The National Commissioner then refers the application to the Critical Infrastructure Council, which reviews the application and the proposed risk category, and makes a recommendation to the Minister of Police regarding the outcome of the application. Based on the Council's recommendation, the Minister can declare infrastructure as critical infrastructure.

According to draft regulations issued in 2023, the National Commissioner of the South African Police Service must maintain a registry of all critical infrastructure and make the registry available to the public on its website.[30] However, the draft regulations do not appear to have been finalized, nor does the South African Police Service appear to have a registry of critical infrastructure on its website. Because of this, it remains unclear whether any infrastructure has yet been designated as critical infrastructure under the Act. Moreover, it is unclear whether South Africa's Koeberg nuclear power plant would be considered critical infrastructure by the relevant entities such that it would be formally designated as critical infrastructure.

Separately, South Africa has a National Nuclear Regulator that develops "regulatory requirements for the assurance of nuclear security or physical protection system[s] (PPS) at nuclear installations or associated actions in the Republic of South Africa."[31] The nuclear sector in South Africa is largely governed by three acts: (1) the Nuclear Energy Act of 1999,[32] (2) the National Nuclear Regulator Act (NNRA) of 1999,[33] and (3) the National Radioactive Waste Disposal Institute Act of 2008.[34] While these three acts work in concert to protect and regulate

[26] Id. at Section 11.
[27] Id. at Section 1.
[28] Id. at Section 16.
[29] See also Critical Infrastructure Protection Regulations (draft), 2023, http://www.policesecretariat.gov.za/downloads/Regulations/Critical_Infrastructure_Protection_Regulations.pdf
[30] Id. at Section 23.
[31] South African National Nuclear Regulator https://nnr.co.za/about/nuclear-security/
[32] https://www.gov.za/documents/nuclear-energy-act
[33] https://www.gov.za/documents/national-nuclear-regulator-act
[34] https://www.gov.za/documents/national-radioactive-waste-disposal-institute-act

nuclear facilities and material, they do not specify that nuclear facilities comprise part of the country's critical infrastructure. Moreover, the NNRA does not specifically require the National Nuclear Regulator to oversee nuclear security; however, it appears that it is doing so in practice.

As South Africa operationalizes its Critical Infrastructure Protection Act, it may wish to consider how designating the Koeberg nuclear power plant as part of the country's critical infrastructure will strengthen its protection and compliment the aims of nuclear security and physical protection.

c. United States

In 2013, the White House released the Presidential Policy Directive (PPD) on Critical Infrastructure Security and Resilience, aiming to "advance[] a national unity of effort to strengthen and maintain secure, functioning, and resilient critical infrastructure."[35] The PPD describes critical infrastructure and resilience as a "shared responsibility among the Federal, state, local, tribal, and territorial (SLTT) entities, and public and private owners and operators of critical infrastructure," although the Federal government has responsibilities to protect its critical infrastructure and ensure continuity of operations, among other areas of responsibility and coordination.[36] The PPD identifies the energy sector as one that is "uniquely critical" due to the "enabling functions" it provides across all critical infrastructure sectors.[37] The U.S. Department of Homeland Security provides strategic guidance and coordination for the protection and resilience of critical infrastructure, while other federal agencies provide sector-specific support. The PPD also provides that the Department of Homeland Security will operate two critical infrastructure centers, one for physical infrastructure and one for cyber infrastructure, to serve as "focal points" for partners to obtain information regarding the protection of critical infrastructure.[38]

The PPD on Critical Infrastructure Security and Resilience defines critical infrastructure as "systems and assets, whether physical or virtual, so vital to the United States that the incapacity or destruction of such systems and assets would have a debilitating impact on security, national economic security, national public health or safety, or any combination of those matters." Together with the National Infrastructure Protection Plan, the Nuclear Sector-Specific Plan details measures to identify and manage risk to nuclear power plants, reactors, fuel-cycle facilities, and nuclear and radioactive materials used for medical, industrial, and academic purposes.[39] The U.S. Department of Homeland Security has also developed the Nuclear Sector Cybersecurity Framework Implementation Guidance for owners and operators in the nuclear sector.[40]

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[35] The White House, Presidential Policy Directive 21, Critical Infrastructure Security and Resilience, Feb. 12, 2013, https://www.cisa.gov/sites/default/files/2023-01/ppd-21-critical-infrastructure-and-resilience-508_0.pdf.
[36] Id.
[37] Id.
[38] Id.
[39] U.S. Department of Homeland Security, "Nuclear Reactors, Materials, and Waste Sector-Specific Plan: An Annex to the NIPP 2013," 2015, https://www.cisa.gov/sites/default/files/publications/nipp-ssp-nuclear-2015-508.pdf.
[40] U.S. Department of Homeland Security, "Nuclear Sector: Cybersecurity Framework Implementation Guidance," May 2020, https://www.cisa.gov/sites/default/files/publications/Nuclear_Sector_Cybersecurity_Framework_Implementation_Guidance_FINAL_508.pdf.

III. Identifying Best Practices of Legal and Regulatory Frameworks on Protecting Critical Infrastructure

a. Discussion

Amongst the various states and regional bodies identified above, each jurisdiction has developed a distinct legal framework for critical infrastructure. These varied approaches are understandable given the unique aspects of a state's infrastructure that could be deemed critical, as well as the fact that the concept of critical infrastructure may encompass a range of sectors, entities, and relevant government bodies or ministries that may need to take actions to protect critical infrastructure and coordinate with a range of entities, including law enforcement, emergency response, and federal, domestic, and local authorities to respond quickly and appropriately to threats to or attacks on critical infrastructure. For these reasons, the legal framework and operational response to threats to critical infrastructure could be broad or different in scope than required for a threat to or attack on nuclear infrastructure.

At the same time, domestic nuclear security regimes typically designate the authority for regulating and inspecting nuclear security at facilities to a specialized nuclear regulatory body. Given the unique technical and legal requirements for nuclear security, the unique risks posed by the unauthorized access to or use of nuclear materials, among other requirements for nuclear facilities and materials, nuclear regulatory bodies would not necessarily have authority over the range of other kinds of infrastructure that may also be designated as "critical." Accordingly, a need for careful coordination across government remains vital, both in terms of protecting critical infrastructure more broadly, and protecting nuclear facilities as a subset of critical infrastructure more specifically.

b. Recommendations

Understanding the varying approaches to the frameworks for critical infrastructure outlined above, states, regional bodies, and international organizations may wish to consider incorporating the following features when developing critical infrastructure frameworks:

- Recommendation: States should consider declaring nuclear reactors and other nuclear facilities part of "critical infrastructure" in their national laws and regulations.

Due to nuclear power's presence as a growing energy supply and the inherent dangers of malicious acts and sabotage at nuclear reactors and other nuclear facilities, as well as growing cybersecurity threats, explicit inclusion of nuclear facilities within the critical infrastructure framework removes any ambiguity and allows for further protections. Even if a country does not have nuclear reactors or other nuclear facilities within its borders, explicitly including nuclear facilities and material in the definition of "critical infrastructure" provides clarity and protection to any future use or application of nuclear power within the state.

- Recommendation: States should develop a comprehensive risk management plan that explicitly outlines actions to take for protecting security at nuclear reactors and other nuclear facilities. States should allocate resources towards these security measures at nuclear facilities.

A risk-based approach, recommended by the IAEA's NSS 24-G, is the basis for the development of nuclear security measures at nuclear facilities and for risk management of critical infrastructure. NSS 24-G describes concepts and methodologies for a risk-informed approach "including identification and assessment of threats, targets, and potential consequences; threat and risk assessment methodologies; and the use of risk informed approaches as the basis for informing the development and implementation of nuclear security systems and measures."[41] Ensuring that nuclear power plants are included as part of a state's critical infrastructure, in addition to using a risk-informed approach, is crucial to creating critical infrastructure security environments that promote sustained coordination and collaboration. It is paramount that national stakeholders with responsibilities in establishing security requirements for nuclear power plants and other stakeholders that have responsibilities for establishing regulatory requirements for other critical infrastructure entities closely communicate and coordinate at the national level. Without close communication and coordination, ambiguity or uncertainty can emerge, which presents a risk to all critical infrastructure, including nuclear power plants.

- Recommendation: States should ensure that critical infrastructure frameworks effectively interface with frameworks for nuclear security, as well as frameworks for nuclear and radiological emergency preparedness and response.

Critical infrastructure frameworks have complimentary goals to frameworks on nuclear and radiological emergency preparedness and response and nuclear security. Critical infrastructure frameworks aim to ensure the resilience and protection of economies and societies. By seeking to protect nuclear facilities and material, nuclear security frameworks such as the Convention on the Physical Protection of Nuclear Material (CPPNM) and its Amendment also seek to protect people and the environment from the effects of theft, smuggling, sabotage, or other harms. To ensure that each framework accomplishes its goals while protecting the appropriate facilities and materials, it will be important to review critical infrastructure frameworks to ensure they compliment any existing requirements related to nuclear security or emergency preparedness and response emanating from international instruments. Cross-border coordination may also be valuable to consider and explore when developing critical infrastructure frameworks.

IV. Conclusion

Any discussions of critical infrastructure within a state should include nuclear reactors and other nuclear facilities. The interconnectedness nature of cross-sector entities, such as energy, transportation systems, communications, emergency services, water, information technologies and others, result in inevitable synergies between legal frameworks for the security of nuclear facilities and legal frameworks for the protection of critical infrastructure.

The protection of nuclear facilities against sabotage and other malicious acts is paramount in ensuring energy security and uninterrupted energy supply. The protection of other sectors, such

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[41] International Atomic Energy Agency, Nuclear Security Series (NSS) No. 24-G, https://www.iaea.org/publications/10677/risk-informed-approach-for-nuclear-security-measures-for-nuclear-and-other-radioactive-material-out-of-regulatory-control

as communications, water supply, and others, also supports a safe and secure operation of nuclear facilities. Some countries rely on broader critical infrastructure frameworks to impose security requirements on nuclear facilities, or to achieve robust cybersecurity systems. The inclusion of nuclear facilities and materials as part of these frameworks helps ensure their overall security and resilience.

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